AMD AI DEVMASTER HACKATHON 2026 / TRACK 2

YIJIAN:
From Superhuman Go to Human Understanding

A private, evidence-grounded AI Go coach powered by KataGo, Qwen and AMD Radeon.

PARTICIPANT	ALI MOQI
DEMO GPU	AMD RADEON PRO W7900 48 GB
TAGLINE	EVERY SUPERHUMAN MOVE, MADE UNDERSTANDABLE.

The gap is not playing strength. It is trustworthy explanation.

Go engines already find superhuman moves. Learners still struggle because a ranked candidate does not explain direction, thickness, territory, initiative, trade-offs or timing. A generic language model can sound persuasive while discussing the wrong move or inventing a board relationship.

YIJIAN treats teaching quality as an agent architecture problem: reconstruct the position, gather authoritative evidence, identify the user's subject, generate an explanation, validate it, and repair it before display.

One review loop

Stage	User-visible outcome
Reconstruct	Exact board state from SGF/GIB, including turn and variation.
Analyze	KataGo candidates, values, visits and principal variations.
Teach	Go concepts selected for the actual position and question.
Validate	Coordinates, regions, subject identity and numbers checked.
Interact	Follow-up questions remain anchored to the selected move.

Every superhuman move, made understandable.

The product is a review workspace, not a chat window.

The board, engine evidence, teaching card, variation controls and follow-up dialogue share one position context. The user can ask about the played move or the AI move without losing the tactical state.



The current interface is Chinese because the initial coaching audience is Chinese-speaking. Competition material is English.

Server authority keeps four kinds of truth separate.

Layer	Responsibility
KataGo facts	Candidate ranking, visits, win rate, score lead and principal variations.
Go-rule facts	Legality, board replay, captures, liberties, adjacency and coordinates.
Teaching heuristics	Thickness, urgency, direction, timing and trade-off interpretations.
LLM expression	Question-appropriate wording and structure, constrained by evidence.

Agent control flow

- FastAPI creates a position context from game, turn, variation and board hash.
- KataGo and deterministic board analysis populate a structured teaching context.
- Qwen3.6-27B receives evidence plus conversation history and writes the explanation.
- A validator checks subject, coordinates, region language and numerical claims.
- Failed answers are repaired with explicit error feedback or replaced by a deterministic fallback.
- SQLite caches evidence and narratives for fast reopening.

The browser never decides which engine fact is true.

The agent follows the question, not a rigid answer template.

Subject resolution

A question can target the played move, the AI recommendation, a named coordinate or a follow-up concept. The server resolves that subject before model generation.

Variation evidence

The system reads multiple KataGo continuations before forming a teaching conclusion, so local fighting, liberties, initiative and follow-up order can be discussed from actual branches.

Adaptive explanation

Opening comparisons, tactical fights, ko questions and endgame choices require different structures. The model selects the shape of the answer while the evidence contract remains fixed.

Consistency repair

A move in the upper-left cannot be described as lower-center. A question about the AI move cannot silently become an explanation of the played move. Contradictions trigger repair.

Deterministic fallback

If the model is unavailable or cannot produce a valid answer, the user still receives engine numbers, subject identity and a safe evidence summary.

Fluency is optional. Evidence consistency is not.

One W7900 runs both the superhuman engine and the local coach.

The demo instance uses an AMD Radeon PRO W7900 48 GB. KataGo v1.16.5 runs with the official b40 network through OpenCL. Qwen3.6-27B Q4_K_M runs in llama.cpp through ROCm 7.2.1.

The GPU schedule avoids maximum KataGo and LLM load at the same time: scan first, verify critical nodes, then generate narratives and cache the result.

Measured workload	Result
33 positions x 32 visits	13.55 seconds total
Critical position x 128 visits	2.63 seconds
Coach constructed acceptance set	6/6 cases passed
Coach acceptance latency	P95 9.536 seconds
Automated regression suite	373 tests

Performance boundary

The six-case coach set is a constructed acceptance benchmark, not a claim of general Go knowledge. Position complexity and contention affect latency. KataGo remains the numerical authority.

Privacy is an explicit capability boundary.

- Game records remain local unless the user explicitly enables remote KataGo.
- Model and engine endpoints bind to loopback interfaces.
- Remote access uses an authenticated SSH tunnel.
- Provider keys stay in server environment variables and never enter browser state.
- The public submission excludes private SGFs, SQLite databases, creator transcripts, review drafts, authorization conversations, model weights and credentials.

Current capability boundary

- Long-term personalized student memory is not yet a production feature.
- Licensed-corpus retrieval is not yet the main answer path.
- The current main runtime uses a 27B base model plus evidence construction and validation; it does not present private creator data as public model training.
- Expert evaluation must expand beyond the small constructed acceptance set.

Trust comes from making boundaries visible.

Why YIJIAN belongs in Track 2

YIJIAN demonstrates an AI agent that plans and executes a multi-step review workflow: tool calling, evidence assembly, local model inference, consistency validation, repair, interactive follow-up and persistent state.

The AMD GPU is not an incidental accelerator. It makes the combined engine-plus-coach experience practical on one professional Radeon card while keeping sensitive Go records under user control.

Submission contents

- English README and source setup guide.
- Public-safe source snapshot with tests and Radeon deployment scripts.
- Performance and validation report.
- 3-5 minute English demo runbook.
- English presentation and project specification PDF.

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Project evidence: included benchmark scripts, acceptance tests and submission documentation. Third-party components retain their original licenses.